

RAG Chatbot - Tài Liệu Kỹ Thuật Chi Tiết

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Chunking - Embedding - Retrieval - Generation

Trường Đại học Khoa học Tự nhiên - ĐHQG Hà Nội
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PHAN 1: CHUNKING (Chia tai lieu thanh chunks)

1.1. Tong quan

He thong co 2 che do chunking:

- A) Rule-based (phase1_chunking.py): Dung cho tai lieu thu tuc nhap hoc - phan tich cau truc PHAN 1-4, B1-B4 bang regex.
- B) LLM-powered (automation_ai_rag.py): Dung Groq LLM de chunk tai lieu moi/khong co cau truc co dinh.

1.2. Che do A: Rule-based Chunking (phase1_chunking.py)

EnrollmentProcessor xu ly tai lieu thu tuc nhap hoc theo cau truc 4 cap:

Level 0: Overview - Tom tat toan bo tai lieu

Level 1: Section - Moi PHAN lon (PHAN 1, 2, 3, 4)

Level 2: Subsection - Nhom thong tin trong moi phan (fee_overview, steps_group, schedule_group...)

Level 3: Detail - Chi tiet cu the (fee_item, step_b1, schedule_major, docs_same_day_item...)

1.2.1. Metadata duoc gan nhu the nao (Rule-based)

Moi chunk duoc gan metadata sau:

- intent_key: Xac dinh theo vi tri trong cau truc (PHAN 1->lookup, PHAN 2->online_confirmation, PHAN 3->fee_info, PHAN 4->admission_procedure)
- subtype: Loai chi tiet (fee_item, step_b1, schedule_major, section_3...)
- section_id: phan_1, phan_2, phan_3, phan_4
- step_id: b1_phan_4, b2_phan_4... (chi co trong PHAN 4)
- canonical_nav_id: ID dieu huong chinh (= section_id hoac step_id)
- year: Nam tuyen sinh tu ten file hoac .env
- source: Ten file nguon
- Cac truong bo sung: dates, links, amount, major, major_norm, time_slot...

1.2.2. Quy tac xac dinh intent_key

PHAN 1 -> lookup (tra cuu trung tuyen)

PHAN 2 -> online_confirmation (xac nhan nhap hoc truc tuyen)

PHAN 3 -> fee_info (hoc phi), fee_payment (hinh thuc nop)

PHAN 4:

- Tong quan/buoc -> admission_procedure
- Tung buoc B1-B4 -> step
- Lich theo nganh -> schedule_by_major
- Ho so nop ngay -> docs_same_day
- Ho so nop sau -> docs_later
- Luu y -> notes
- Lien he -> contact
- Moc thoi gian -> deadlines_summary

1.3. Che do B: LLM Chunking (automation_ai_rag.py)

Dung khi tai lieu khong co cau truc PHAN 1-4 co dinh.

Goi Groq API voi prompt yeu cau LLM chia tai lieu thanh Parent/Children.

1.3.1. Prompt gui den LLM de chunk

System prompt (tom tat):

"Ban la chuyen gia boc tach du lieu tuyen sinh.

Nhiem vu: Chia tai lieu thanh cac PHAN LON (Parents)

va cac Y NHO CHI TIET (Children).

Tap hop nhan (topics): [fee, admission_docs, schedule,

procedure, contact, major_info, general, fee_payment]

QUY TAC PHAN CHIA (BAT BUOC):

1. GIU NOI DUNG O PARENT: Parent chunk PHAI chua

DAY DU noi dung goc.

2. KHONG CHIA QUA NHO: Gom nhom thong tin lien quan.

3. BOC TACH LICH: Gom nhom lich theo NGAY.

4. CHI TIET GIAY TO: Khong chia moi giay to thanh 1 Child.

5. KHONG DUOC BO SOT: Quet het toan bo van ban.

Yeu cau dinh dang JSON:

```
{ sections: [{ parent_title, parent_content, topic,
children: [{ child_title, child_content, keywords }] }] }
```

1.3.2. LLM du doan metadata nhu the nao

LLM tra ve JSON voi cac truong:

- parent_title: Tieu de muc lon
- parent_content: Noi dung goc day du
- topic: Nhan phan loai (fee, schedule, procedure...)
- children[].child_title: Tieu de y nho
- children[].child_content: Noi dung chi tiet
- children[].keywords: Tu khoa

Code sau do gan them:

- chunk_id (tu dong)
- level: parent/child
- source: ten file
- year: tu ADMISSION_YEAR

PHAN 2: EMBEDDING (Tao vector va luu tru)

2.1. Model embedding

Model: Gemini Embedding API (gemini-embedding-001)

Goi qua REST API:

POST https://generativelanguage.googleapis.com/v1beta/models/gemini-embedding-001:embedContent?key={GEMINI_API_KEY}

Khong dung SDK, chi dung thu vien requests.

Cau hinh trong .env: GEMINI_MODEL_EMBED=gemini-embedding-001

2.2. Chuan bi text truoc khi embed

Ham `prepare_text_for_embedding(chunk)` tao text gom:

- Dong 1: [TYPE: ...] [YEAR: ...] title
- Dong 2+: Noi dung chunk
- Dong cuoi: Metadata bo sung (total, date, major, nav_id, section, step)

Muc dich: Giup embedding bao gom ca ngu canh metadata, khong chi noi dung thuan.

2.3. Luu tru vector

File: `vector_db.json`

Cau truc: { "chunks": [...], "embeddings": [...] }

Moi chunk tuong ung 1 embedding vector cung index.

Truy van: Cosine similarity voi NumPy:

`similarities = dot(E_normalized, query_normalized)`

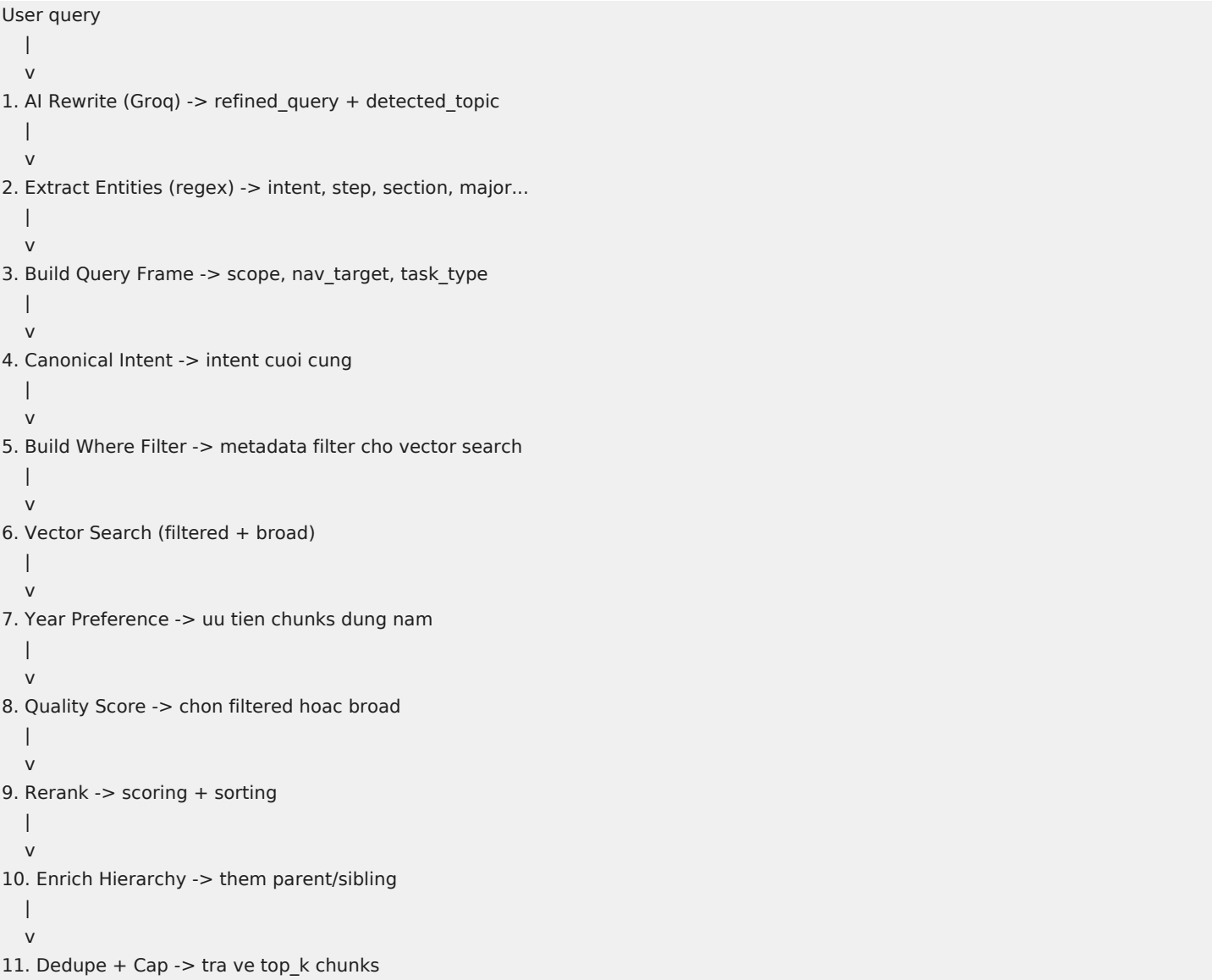
Loc theo where filter (metadata match)

Sap xep giam dan similarity

Tra ve top-k ket qua

PHAN 3: TRUY VAN (Retrieval Pipeline)

3.1. Tong quan luong truy van



3.2. Co du doan intent tu query khong?

CO. He thong du doan intent qua 3 nguon:

- 1) AI topic detection: Groq LLM phan loai query thanh topic (fee, schedule, procedure...)
- 2) Rule-based entity extraction: Regex phat hien tu khoa (hoc phi, lich, buoc, ho so...)
- 3) Canonical intent mapping: Ket hop AI topic + entities + query_frame de xac dinh intent cuoi cung.

Thu tu uu tien:

- Query frame nav_type (local_step -> step, section -> admission_procedure)
- Entity keywords (hoc phi -> fee_info, lich -> schedule_by_major...)
- AI topic fallback (neu khong match rule nao)

3.3. Truy van theo embedding (Vector Search)

Quy trình day du:

1. Tao embedding cho refined_query (Gemini API)
2. Chay 2 luong song song:
 - a) Filtered search: query + where_filter (metadata match)
 - b) Broad search: query + khong filter
3. Tinh quality score cho moi luong (top-3 grounding score trung binh)
4. Chon luong tot hon:
 - Dung broad neu: filtered rong HOAC broad_score > filtered_score + 0.10
 - NHUNG: neu strict nav lock (section/local_step) -> giu filtered
5. Rerank ket qua da chon

3.4. Truy van theo luat cung (Metadata Filter)

Where filter duoc xay dung tu intent + entities:

- Nav target la buoc local (vd: b2_phan_4):
 - > where = {canonical_nav_id: 'b2_phan_4'}
- Nav target la section (vd: phan_3):
 - > where = {section_id: 'phan_3'}
- Intent cu the (fee_info, step, schedule_by_major...):
 - > where = {intent_key: intent} + subtype/step_id bo sung
- Intent chung (general, admission_procedure):
 - > where = None (khong filter, dung embedding similarity)

3.5. Rerank scoring formula

Cong thuc:

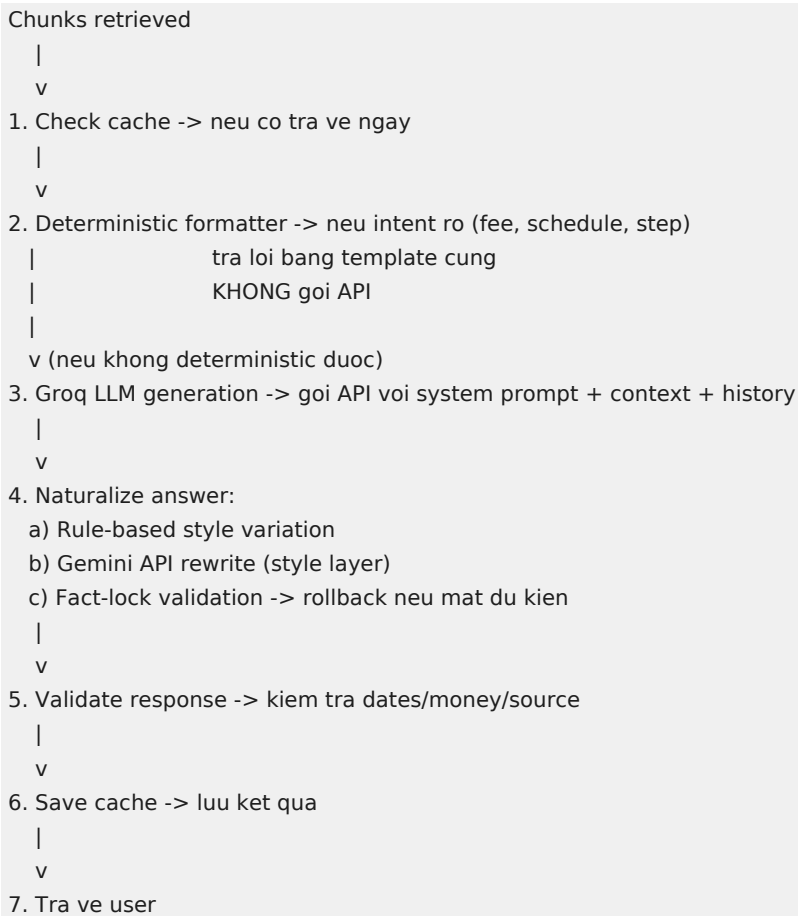
$$\text{grounding_score} = (\text{base} * 0.70) + (\text{lexical_score} * 0.30) + \text{bonus}$$

Trong do:

- base = cosine similarity tu vector search
- lexical_score = (overlap_query * 0.6) + (overlap_refined * 0.4)
(ty le token trung giua query va noi dung chunk)
- bonus:
 - +0.15 neu intent_key khop
 - +0.25 neu dung step number
 - +0.30 neu dung major (nganh)
 - +0.20 neu dung docs subtype
 - +0.20 neu dung nam tuyen sinh (ADMISSION_YEAR)

PHAN 4: GENERATION (Tao cau tra loi)

4.1. Pipeline generation



4.2. Deterministic formatter

Xu ly cac intent co cau truc ro rang ma KHONG can goi LLM:

- fee_info: Liet ke tung khoan phi + tinh tong
- fee_payment: Hinh thuc va thoi gian nop
- admission_procedure: Tom tat PHAN 1-4 hoac buoc B1-B4
- step: Noi dung cu the cua buoc Bn
- schedule_by_major: Lich theo nganh
- docs_same_day: Ho so nop trong ngay
- docs_later: Ho so nop sau

Uu diem: Nhanh (0ms), chinh xac 100%, khong ton quota API.

4.3. LLM Generation (Groq)

Khi deterministic khong xu ly duoc, goi Groq API:

System prompt yeu cau:

- Tra loi DUA HOAN TOAN tren context

- Không tự suy diễn
- Tính toán chính xác số liệu
- Trích dẫn nguồn [SOURCE]
- Trả về JSON: {answer, used_chunk_ids, grounded, uncertainty_note}

Messages gửi đi:

- system: system_prompt + answer_contract
- user: [CHAT_HISTORY] + [CONTEXT] + query

4.4. Naturalization (Làm tự nhiên câu trả lời)

2 lớp naturalization:

Lớp 1 - Rule-based (_safe_style_variation):

- Thay đổi header theo style (friendly/formal/concise)
- VD: 'Tom tắt thu tục...' -> 'Minh tóm tắt...' (friendly)
- Fact-lock: rollback nếu mất ngay/tiền/URL/PHÂN/B1-B4/[SOURCE]

Lớp 2 - Gemini API (_call_gemini_naturalizer):

- Gửi câu trả lời đã rule-based cho Gemini viết lại văn phong
- Prompt: 'Chỉ thay đổi văn phong, KHÔNG thay đổi nội dung cốt lõi'
- Temperature: 0.2 (rất thấp để giảm sáng tạo)
- Fact-lock validation: kiểm tra locked tokens trước/sau
- Nếu mất bất kỳ token nào -> rollback về bản rule-based
- Nếu thêm token mới (ngay/tiền/URL) -> rollback
- Có thể tắt: NATURALIZE_ENABLED=false

PHAN 5: QUAN LY HISTORY (Lich su hoi thoai)

5.1. Luu tru history

- Luu trong memory (dict): chat_sessions[user_id::session_id]
- Format: List[{role: 'user'/'assistant', content: '...'}]
- Gioi han: MAX_HISTORY_TURNS = 20 luot gan nhat
- Reset: Xoa khi user bam reset hoac doi session

5.2. History duoc dung o dau?

1) RETRIEVER (automation_retriever.py):

- Giai quyet follow-up: 'buoc tiep theo' -> xem history de biet dang o buoc nao
- _infer_relative_axis_from_history: xac dinh dang noi global (PHAN) hay local (B)
- _last_step_from_history: tim buoc cuoi cung da tra loi
- _last_section_from_history: tim phan cuoi cung da tra loi

2) GENERATOR (phase5_llm_generation.py):

- _build_history_context: Dong goi 10 luot gan nhat thanh text
- Dinh dang:
[CHAT_HISTORY]
USER: cau hoi cu
ASSISTANT: cau tra loi cu
[END_CHAT_HISTORY]
- Ghep vao dau prompt gui cho LLM

5.3. Prompt cuoi cung gui cho LLM gom nhung gi?

Thu tu trong messages:

1. System message:

- System prompt (vai tro, nguyen tac tra loi)
- Answer contract (yeu cau JSON output)

2. User message (ghep theo thu tu):

- [CHAT_HISTORY]...10 luot gan nhat...[END_CHAT_HISTORY]
- [CONTEXT]...chunks retrieved...[END CONTEXT]
- Cau hoi hien tai cua user

Nhu vay LLM nhan duoc: history + context tu vector DB + query moi de tra loi co ngu canh day du.

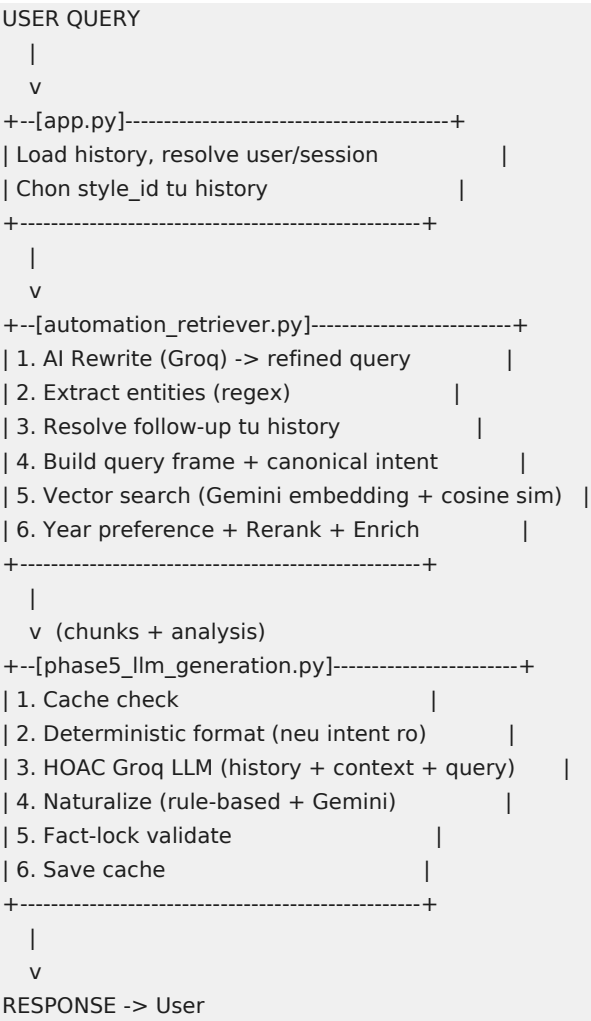
5.4. Style rotation theo history

He thong tu dong xoay van phong tra loi:

- 3 style: formal, friendly, concise

- Xoay theo so luot assistant da tra loi + hash(session_key)
- Moi luot tra loi co style khac -> tranh lap di lap lai
- Variant suffix (_v0, _v1, _v2) them da dang cau van

PHAN 6: SO DO TONG THE



External APIs su dung

- 1. Gemini API:
 - Embedding (gemini-embedding-001): Tao vector cho chunks va query
 - Naturalizer (gemini-1.5-flash): Viet lai van phong cau tra loi
- 2. Groq API:
 - Query rewrite: Tinh chinh cau hoi user
 - Topic detection: Phan loai chu de
 - LLM Generation: Tao cau tra loi khi deterministic khong du
 - AI Chunking: Chunk tai lieu moi (automation mode)

Fallback khi API loi

- 4 lop fallback:
 - 1. Deterministic formatter (khong can API)
 - 2. Response cache (cau tra loi cu da luu)
 - 3. Template response (cau tra loi mau theo intent)
 - 4. Thong bao loi than thien